Hotel Booking Demand Clustering Study

A Reproducible Experimental Study of Clustering

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Abstract

This report presents a reproducible clustering study of hotel booking demand data with the goal of partitioning bookings into interpretable guest segments based solely on pre-stay observable features. Two representations were studied: R-EUCLID-robust-weighted-withADR and a pipeline that excludes the daily rate R-EUCLID-robust-weighted-noADR. K-Means ($k = 3$, Silhouette = 0.452, ARI = 0.989 across 10 seeds), iK-Means (automatic $k = 8$), and Agglomerative Hierarchical clustering (Ward, Silhouette = 0.453 at $k = 3$) are compared. The selected three-cluster solution identifies a Non-Refundable Advance Booker segment (12.4%), a Standard Flexible City Guest segment (70.7%), and a Premium Resort Long-Stay segment (16.8%). The noADR comparison confirms that the commitment axis (`is_non_refund`) drives cluster separation independently of price.

Contents

1 Introduction

1.1 Context

Hotel revenue management depends on understanding who books, when they book, and how their bookings differ in value and service requirements. Booking records contain pre-arrival signals such as rate, stay duration, booking horizon, deposit commitment, meal plan, parking need, and special requests. These signals can reveal latent booking types that support pricing, upselling, and operational planning decisions.

1.2 Segmentation question

The concrete segmentation question addressed in this project is:

> Can hotel booking records be partitioned into distinct and interpretable behavioural segments using only pre-stay observable characteristics, such as daily rate, total estimated spend, length of stay, booking horizon, meal plan, deposit commitment, special requests, and parking need, in order to identify guest profiles that support targeted revenue-management and personalised-service strategies?

The unit of analysis is the individual booking record. A guest may appear more than once in the dataset, so the clusters should be interpreted as booking segments rather than permanent guest identities.

1.3 Why clustering is appropriate

In this project, we don't know beforehand which group each customer belongs to. Our goal is to discover these groups rather than making predictions. Therefore, unsupervised techniques, like clustering, are the right choice.

Clustering is ideal here because hotel bookings have very different characteristics from one another: some people book far in advance, others the day before, there are differences in prices, length of stay, type of deposit paid, and special requests.

1.4 Research questions

The report addresses the course research questions as follows:

- **RQ1 – Problem framing:** define a booking-level segmentation question and justify clustering.
- **RQ2 – Representation:** define leakage-safe preprocessing and feature representations for mixed-type booking data.
- **RQ3 – Modelling:** compare K-Means, iK-Means, and Ward hierarchical clustering.
- **RQ4 – Interpretation:** profile the discovered clusters as booking segments.
- **RQ5 – Robustness:** evaluate sensitivity to seeds, subsampling, scaling, feature weighting, and representation changes.

2 Dataset and Preprocessing

2.1 Dataset documentation

Table 1: Dataset documentation and traceability.

Item	Details
Dataset	Hotel Booking Demand
Version	Course release v1
Unit of analysis	One row corresponds to one booking record
Original size	119,390 rows and 32 columns
Time span	Arrivals from July 2015 to August 2017
Source	António, de Almeida, and Nunes (2019); dataset provided through the course platform
Raw-data governance	Raw CSV is not committed to the repository; it must be placed in `data/`
Checksum	SHA-256: `7c2ae42a7353905ea136e5c2287f17c92c5435826598bfbb8491c6f0c7b1fc06`
Usage terms	Dataset used under the course release terms, no raw data redistributed in the repository

2.2 Data quality

The main data-quality issues identified are:

- `children` has 4 missing values, imputed as 0 because missing children counts are interpreted as no children in the booking.
- `country` has 488 missing values and is retained only for post-hoc profiling, not as a clustering input.
- `agent` and `company` have high missingness and high cardinality; both are dropped as ID-like fields.
- `adr` has a heavy right tail (maximum approximately €5,400) and one negative value, requiring robust preprocessing and careful interpretation.
- Rows with `adr = 0` are excluded from clustering because they correspond to complementary or test-like bookings. This removes 1,960 records, leaving 117,430 analytic records.

Table 2: Main missingness and governance decisions.

Variable	Missing	Role	Decision
`children`	4	Numerical	Impute as 0
`country`	488	Categorical	Exclude from clustering; retain for profiling
`agent`	16,340	ID-like categorical	Drop due to high missingness/cardinality
`company`	112,593	ID-like categorical	Drop due to high missingness/cardinality

2.3 Leakage control

The segmentation index time is the booking creation time. Therefore, only variables available at or before booking time can be used to form clusters. The following variables are excluded from clustering inputs:

- `is_canceled`: outcome variable.
- `reservation_status`: post-decision/post-arrival status.
- `reservation_status_date`: date associated with the realised status.
- `agent` and `company`: high-cardinality ID-like fields.
- `country`: excluded from clustering to avoid geographical profiling effects; retained only for post-hoc descriptive analysis.

2.4 Feature engineering

The final clustering features are:

- `total_nights` = `stays_in_week_nights` + `stays_in_weekend_nights`.
- `total_spend_log` = $\log(1 + \texttt{adr}) \times$ `total_nights`, used as a compressed estimated-spend proxy.
- `is_non_refund` = 1 if `deposit_type` is Non Refund, 0 otherwise.
- `lead_time_rank`, an ordinal quantile binning of `lead_time`: 0 for last-minute, 1 for mid-horizon, and 2 for advance bookings.
- `meal_encoded`, an ordinal encoding of meal-plan inclusivity: SC/Undefined = 0, BB = 1, HB = 2, FB = 3.
- `total_of_special_requests`.
- `required_car_parking_spaces`.

2.5 Representations and distance

Two Euclidean representations are studied:

Table 3: Final representations.

Representation	**Definition**
R-EUCLID-robust-weighted-withADR	**Features:** `total_spend_log`, `total_nights`, `is_non_refund`, `total_of_special_requests`, `required_car_parking_spaces`, `meal_encoded`, `lead_time_rank`. **Scaling/weights:** RobustScaler on continuous variables; `is_non_refund` ×3.0; `meal_encoded` ×0.5.
R-EUCLID-robust-weighted-noADR	**Features:** same as R-EUCLID-robust-weighted-withADR, excluding `total_spend_log`. **Scaling/weights:** same scaling and weights as R-EUCLID-robust-weighted-withADR.

All models and internal validity indices are computed in the same weighted Euclidean space used to fit the corresponding clustering model.

3 Clustering Methods

3.1 K-Means baseline

K-Means is used as the mandatory baseline. The predefined search interval is $k \in \{2, \ldots, 10\}$, chosen because $k = 2$ is the minimum meaningful segmentation and values above 10 are unlikely to remain managerially interpretable. The configuration is `n_init=10`, `max_iter=300`, and `random_state=42` for the main run (Runtime: ∼5 seconds).

3.2 iK-Means

Intelligent K-Means (iK-Means) was included as an exploratory method to automatically discover cluster centers. Using Mirkin's procedure with `min_cluster_size=1000`, the algorithm automatically extracted $k = 8$ clusters (Runtime: $\sim$3 seconds). However, iK-Means was not selected as the primary segmentation model because 8 clusters overly fragment the main booking profiles, adding complexity without offering qualitatively new, actionable insights for revenue management.

3.3 Agglomerative hierarchical clustering

Agglomerative hierarchical clustering with Ward linkage is used as the alternative clustering family. Ward linkage is compatible with the chosen representation because the features are embedded in a Euclidean vector space. Due to computational cost, Ward clustering is run on a reproducible subsample of 30,000 records. The main subsample uses seed 42, and robustness is assessed using additional subsamples.

4 Experimental Protocol

4.1 Internal validity

Models are evaluated using:

- **Silhouette score** as the primary compactness/separation index.
- **Calinski-Harabasz index** as an additional compactness/separation index, where higher is better.
- **Davies-Bouldin index**, where lower is better.

4.2 Family-specific diagnostics

For hierarchical clustering, a Ward dendrogram is used to inspect merge structure and support the choice of the cut. For iK-Means, the automatic number of clusters and any anomalous extracted clusters should be documented.

4.3 Stability and sensitivity

The robustness protocol includes:

- K-Means repeated across 10 random seeds for $k \in \{2, 3, 4, 5\}$.
- Adjusted Rand Index (ARI) between runs as the stability measure.
- Hierarchical clustering repeated on controlled subsamples.
- RobustScaler versus StandardScaler as a preprocessing sensitivity check.
- withADR versus noADR comparison to test whether clusters are driven only by price.

4.4 Reproducibility

All experiments are intended to be reproducible from the repository using:

- `experiments.csv` for experiment monitorability (sample_rule, representation_id, model, parameters, runtime, seed, silhouette, calinski, davies, diagnostics, notes).
- `requirements.txt` file for explicit dependency management.
- `run_all.py` as the single-entry point, which explicitly reproduces the entire data mining pipeline (preprocessing, modelling, logging, and figures) from scratch.

5 Results

5.1 Data preparation outcomes

After excluding zero-ADR records, the analytic dataset contains 117,430 bookings. The withADR matrix has 7 features and the noADR matrix has 6 features.

5.2 Model comparison

Table 4: Main model comparison: quantitative results.

Method	Setting	Sil.	CH	DB	Stability
K-Means	$k = 3$, withADR	0.452	63,837	0.857	ARI 0.989 ± 0.007
K-Means	$k = 2$, withADR	0.392	54,947	1.074	ARI 0.998 ± 0.002
K-Means	$k = 3$, noADR	0.443	60,587	0.928	n/a
iK-Means	$k = 8$ (auto)	0.327	56,165	1.012	n/a
Ward	$k = 3$, withADR, $n = 30{,}000$	0.453	15,928	0.832	Sil. 0.445 ± 0.014

Table 5: Main model comparison: interpretation.

Model / setting	Observation
K-Means, $k = 3$, withADR	Selected solution. It gives the best balance between internal validity, seed stability, and business interpretability.
K-Means, $k = 2$, withADR	Stable but too coarse: it merges the Premium Resort Long-Stay profile with the Standard Flexible City Guest profile and loses useful granularity.
K-Means, $k = 3$, noADR	Useful sensitivity check. The noADR solution separates a Special Requests group, a Standard Flexible group, and a Non-Refundable high-lead-time group, confirming that the commitment axis remains visible without price.
iK-Means	Automatic $k = 8$ fragments the main groups into smaller clusters. It is therefore useful as a diagnostic but is not selected as the primary segmentation because it does not reveal qualitatively new booking profiles.
Ward, $k = 3$, withADR	Conceptually distinct method that corroborates the same three-cluster structure on a reproducible subsample.

The selected solution is K-Means with $k = 3$ on the withADR representation. This choice is justified because it achieves the strongest balance between internal validity, stability, and interpretability. Although $k = 2$ is stable, it merges two managerially different segments. Values above $k = 3$ tend to fragment the same business profiles without producing clearer narratives. iK-Means is not selected as the primary solution because its automatic $k = 8$ solution fragments the three main groups without revealing qualitatively new profiles.

5.3 Cluster profiles

Table 6: Selected K-Means cluster profiles for $k = 3$ withADR.

Cluster	Size	%	Main characteristics	Label
C2	14,513	12.4	ADR approximately €90, `is_non_refund`=1.0, stay 2.7 nights, advance lead-time profile	Non-Refundable Advance Booker
C1	83,139	70.7	ADR approximately €103, flexible deposit, stay 2.6 nights, last-minute/mid lead-time profile, highest parking need	Standard Flexible City Guest
C0	19,778	16.8	ADR approximately €114, stay 7.7 nights, total spend 136% above the mean, most special requests, resort-hotel dominant	Premium Resort Long-Stay

5.4 Cluster interpretation

The **Non-Refundable Advance Booker** segment is defined by full deposit commitment (`is_non_refund`= 1.0 for all members), a more advanced lead_time, and short stays of about 2.7 nights.

The **Standard Flexible City Guest** segment forms the majority of bookings (70.7%). These records are typically flexible, short-stay bookings with no non-refundable commitment, an average stay of about 2.6 nights, and the highest parking-need proportion. This segment likely contains both business and urban leisure demand.

The **Premium Resort Long-Stay** segment is the smallest but most valuable profile. It is characterised by longer stays of about 7.7 nights, ADR around €114, total estimated spend 136% above the grand mean, and more special requests. This segment is suitable for pre-arrival personalisation and upselling.

5.5 noADR comparison

The noADR representation tests whether clusters are merely price bands. Removing `total_spend_log` reduces the explicit price/value signal. The Non-Refundable segment remains structurally meaningful, indicating that deposit commitment is an independent axis of segmentation. However, separation between Standard and Premium bookings becomes weaker because price and total spend are important for distinguishing high-value long-stay behaviour.

5.6 Extension E4: PCA Study

The first principal component explains 54.3% of the variance and the first four components together account for 97.2%, indicating that the withADR representation has moderate linear structure. The PCA loadings show that PC1 is dominated by `total_nights`, while PC2 captures the commitment axis (`is_non_refund`).

Table 7: PCA clustering comparison for K-Means $k = 3$ (3 seeds: 42, 7, 123).

Representation	Var. explained	Mean Sil.	Std Sil.	Mean ARI	Std ARI
Original 7D	1.000	0.453	0.002	1.000	0.000
PCA-2	0.773	0.593	0.002	0.992	0.002
PCA-3	0.883	0.504	0.002	0.988	0.004
PCA-4	0.972	0.472	0.003	0.988	0.001

PCA-2 raises the Silhouette to 0.593 and remains highly consistent with the original partition (ARI = 0.992). However, the original 7D representation is retained as the primary space because the PCA axes are linear combinations of the original variables, which reduces direct interpretability for business profiling.

5.7 Extension E1: Cluster-aware Anomaly Analysis

Anomalies are defined relative to the learned cluster structure rather than globally. For each booking i assigned to cluster c_i, the anomaly score is the normalised distance to the centroid:

$$\text{score}_i = \frac{\|x_i - \mu_{c_i}\|_2}{\sigma_{c_i}},$$

where σ_{c_i} is the standard deviation of intra-cluster distances. A high score indicates a booking that is unusual relative to its own segment.

Table 8: E1 anomaly analysis summary.

Metric	withADR
Data-quality suspects (top-20)	18
Rare-but-plausible (top-20)	2
Top-20 overlap (withADR vs noADR)	17/20
Top-20 in all 3 seeds (stability)	20/20

Most data-quality suspects correspond to extreme or inconsistent booking values, especially unusually large ADR or stay-duration combinations relative to their assigned cluster.

Stability (3 seeds: 42, 7, 123): 20/20 anomalies appear in all three top-20 sets, confirming that the anomaly ranking is perfectly robust to K-Means initialisation.

Sensitivity: comparing the withADR and noADR top-20 sets yields an overlap of 17/20, indicating that removing the price signal changes which bookings appear most anomalous. Some outliers are driven by extreme ADR values; without this feature, the commitment and stay-length axes become relatively more important for identifying unusual records.

Limitations: Without true anomaly labels, an unusual booking is not necessarily a data error. Additionally, our anomaly score assumes that clusters are perfectly round, just like K-Means does.

6 Discussion

6.1 Which result best answers the segmentation question?

K-Means with $k = 3$ on the withADR representation best answers the segmentation question. It provides three stable and interpretable booking profiles that align with actionable revenue-management decisions: commitment-based rate locking, flexible city-hotel demand management, and high-value resort-stay personalisation.

6.2 Influence of preprocessing

The feature weight on `is_non_refund` is one of the most consequential preprocessing choices. Since it is a binary variable, its natural range is narrower than the continuous features; however, deposit commitment is assumed to be a key axis for differentiating booking tiers and should not be underweighted relative to price or stay duration. A multiplier of 3.0 was therefore applied, with the specific value determined by experimentation. The meal-plan weight of 0.5 reduces the influence of an ordinal encoding that, while useful for ordering, is less decisive for defining pricing tiers. These weights make the Euclidean metric more aligned with the business segmentation question, but they also introduce subjective assumptions.

6.3 Stability and sensitivity

K-Means is highly stable across seeds for the selected $k = 3$ solution. Hierarchical Ward clustering provides independent support for the three-cluster structure. Sensitivity checks using scaler changes and the noADR representation indicate that the main qualitative conclusions are not driven solely by a single preprocessing decision, although the distinction between Standard and Premium segments depends partly on value-related features.

6.4 Decision-relevant recommendations

- **Non-Refundable Advance Booker:** offer early-bird packages and prepaid upgrades at booking time.
- **Standard Flexible City Guest:** use targeted flexibility-based promotions, parking-inclusive packages, and last-minute offers.
- **Premium Resort Long-Stay:** prioritise pre-arrival personalisation, room upgrades, spa/F&B offers, and long-stay service planning.

6.5 What should not be concluded

The clusters are not externally validated customer types. They are booking-level segments, not permanent guest identities. A returning guest may appear in different clusters across different stays. Internal validity indices measure geometric compactness and separation, not business profitability or causal effects.

7 Limitations, Ethics, and Reproducibility

7.1 Limitations

- There are no ground-truth segment labels.
- The unit of analysis is a booking record, not a unique guest.
- The meal-plan encoding assumes an ordinal structure and equal spacing between meal levels.
- Lead-time quantile binning simplifies a highly skewed continuous variable.
- Manual feature weights encode subjective modelling assumptions.
- The dataset concerns two Portuguese hotels from 2015–2017, limiting generalisability.
- iK-Means produced an automatic $k = 8$ solution, but this result should be treated as exploratory because it fragments the main segments.

7.2 Ethics and governance

The dataset is used according to the course release terms. Raw data are not committed to the repository. No direct personal identifiers are used.

7.3 Reproducibility checklist

The final repository contains:

- `requirements.txt` for explicit dependency management.
- `run_all.py` as the single-entry point to execute the pipeline from scratch.
- `experiments.csv` as the experiment log, recording methods, parameters, runtimes, and metrics.
- Figures and tables are generated dynamically and displayed directly within the Jupyter Notebook upon execution.

7.4 Academic integrity disclosure

Generative-AI tools were consulted only for methodological discussion, syntax review, and report-structure support.

8 Conclusion

This study groups 117,430 hotel bookings based on information available before the guests' stay. Our main model, K-Means with 3 clusters, successfully identified three clear guest profiles: Non-Refundable Advance Booker, Standard Flexible City Guest, and Premium Resort Long-Stay. These results are reliable, as confirmed by our internal metrics, stability tests, and comparison with a hierarchical clustering method. Comparing models with and without the room price (ADR) showed that guest commitment (such as lead time and deposits) is always a key factor. However, including the price provides better insights for revenue management. As a next step, we considered that this analysis should be applied to broader datasets, including more hotels and wider date ranges, to confirm if these guest profiles remain consistent.

A Feature Table

Table 9: withADR feature table.

Feature	Type	Scaling	Weight
`total_spend_log`	Continuous	RobustScaler	$\times 1.0$
`total_nights`	Continuous	RobustScaler	$\times 1.0$
`is_non_refund`	Binary	Raw	$\times 3.0$
`total_of_special_requests`	Count	Raw	$\times 1.0$
`required_car_parking_spaces`	Binary/count	Raw	$\times 1.0$
`meal_encoded`	Ordinal	Raw	$\times 0.5$
`lead_time_rank`	Ordinal	Raw	$\times 1.0$

B K-Means Stability Summary

Table 10: K-Means stability across 10 seeds.

k	Mean ARI	Std ARI	Mean Sil.	Std Sil.	Notes
2	0.9975	0.0021	0.388	0.004	Stable but under-segmented
3	0.989	0.007	0.450	0.002	Selected solution
4	0.9942	0.0057	0.375	0.003	Additional fragmentation
5	0.9803	0.0138	0.327	0.002	Less interpretable

C Scaler Sensitivity

Table 11: Scaler sensitivity for numerical features, $k = 3$.

Scaler	Silhouette	CH	DB
RobustScaler	0.450	63,837	0.857
StandardScaler	0.443	62,062	0.906

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